MATH 1120
CALCULUS II: TEST ONE

Given Time: 2:00 hrs.
March 07, 2026

Name: _____ Student ID: _____

Breakout Session _____

Instructions

- For Part I, give the most simplified answers on the space provided.
- For Part II, show all necessary steps clearly. Answers without supporting work will not receive full credit.
- If additional space is required, use the back side of the paper.
- Sketches must be properly labeled.

Part I: Short Answer Questions (40 Points)

1. What is the average value of the function $f(x) = 2x + 1$ on the interval $[0, 2]$?

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2. If the average value of a continuous function $f(x)$ on the interval $[2, 6]$ is 10, what is the value of $\int_2^6 f(x)\,dx$? _____

3. What is the total mass of a rod of length $2\,m$ whose linear density is $\rho(x) = 3 + 2x + \frac{1}{3}x^2$? _____

4. What is the arc length of the curve $y = \ln(\cos x)$ from $x = 0$ to $x = \dfrac{\pi}{4}$? _____

5. What is an integral representing the area of a surface of revolution when a curve $y = f(x), a \leq x \leq b$, is rotated about the x-axis?

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6. Find the coordinates of the centroid of the region bounded by

$$y = \frac{1}{x}, \quad y = 0, \quad x = 1, \quad x = 3.$$

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7. What is the area of the region between

$$f(x) = 3x^2 + 3 \quad \text{and} \quad g(x) = 4x \text{ over the interval } [0, 1].$$

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8. What is the area of the region enclosed by

$$y = 9 - x^2 \quad \text{and} \quad y = x^2 + 1.$$

Part II: Workout Problems (60 Points)

9. Find the volume of the solid obtained by revolving the region bounded by

$$y = x^2 + 1, \quad y = 0, \quad 0 \leq x \leq 1$$

about the x–axis.

10. Let R be the region in the first quadrant bounded by

$$y = x^3 \quad \text{and} \quad y = 4x - x^2.$$

Find:

(a) the area of R,

(b) the volume obtained by rotating R about the x–axis,

(c) the volume obtained by rotating R about the y–axis.

11. Find the volume of the solid obtained by revolving the region enclosed by

$$y = 5 - x, \quad y = x - 1, \quad x = 0$$

about the y–axis using cylindrical shells.

12. A hemispherical tank with a radius of 3 meters is full of water (density $\rho = 1000\, kg/m^3$). Find the work required to pump all the water out of the top of the tank. (Use $g \approx 9.8\, m/s^2$).

13. (Bonus: 3 pts) Find the surface area generated by rotating the semicircle

$$x^2 + y^2 = 1, \quad y \geq 0$$

about the x–axis.